F1 Podium Prediction — Milestone 1 Analytical Report

Overview

Trained only on seasons up to 2017, our final model, XGBoost, ranks the podium finishers of the unseen 2022–2024 seasons with a test PR-AUC of 0.706 and ROC-AUC of 0.938, against 0.647 PR-AUC for a grid-only baseline. It uses only information known after qualifying and before the race starts.

- **Task:** for every driver in every race, predict whether they finish on the podium (`positionOrder <= 3`).
- **Data:** the Kaggle / Ergast *Formula 1 World Championship (1950–2024)* dataset, 14 relational tables, joined into one modeling table of 24,921 driver-race rows from 1,114 races (1950–2024).
- **Cutoff:** after qualifying, before the start. Current-race results, status, points, laps, pit stops and post-race standings are never features.
- **Base rate:** 13.5% of rows are podiums after cleaning, so PR-AUC is the primary metric.
- **Models:** logistic regression, random forest, XGBoost and a shallow neural network, all on the same rows and 26 features, split by season (train 1950–2017, validation 2018–2021, test 2022–2024). On test they score 0.685–0.724 PR-AUC, within noise of each other; starting position carries most of the signal.

The full analysis, with every plot and experiment, is in the run-all notebook `milestone1Notebook.ipynb`. Section numbers in this report refer to that notebook.

Data cleaning and grain

After cleaning, `results` has exactly one row per (raceId, driverId), every key reconciles, and the podium rate rises from 12.7% to 13.5% because the removed rows were almost never podiums. Raw CSVs are never modified; all cleaning runs on copies (notebook sections 1–2).

Step	What we did	Why
Sentinels (2.1)	Replaced the text `\N` with real missing values in all 14 tables	pandas does not treat `\N` as missing

Step	What we did	Why
Semantic typing (2.2)	Dates to `datetime64`, times of day to `datetime.time`, lap and race times to float seconds; gaps such as `+5.478` added to the winner's time	Durations become comparable numbers; converted finish times match `milliseconds` in 7,651 of 7,680 rows
Keys (2.3)	Checked every primary key (unique, non-null) and 23 foreign-key links	All primary keys valid, zero orphan rows
Indianapolis 500 (2.4)	Removed the 11 Indy 500 races (1950–1960) and subtracted their points from the standings	Only 3 of its 107 drivers raced a normal Grand Prix; 99.8% American entrants; about 37 cars per race against 22
Failed to qualify (2.5.2)	Removed 1,368 `F` rows	These drivers never started; none was ever a podium
Grid 0 (2.5.2)	`grid_numeric` places them behind the last starter; `non_standard_start` flags them	Left as 0 they would look ahead of pole
Mostly empty columns (2.5)	Dropped session dates and times, post-race timing, `q2` / `q3`, driver `number` / `code`, and `pit_stops.duration`	44–99% missing (duration is an exact copy of `milliseconds`); a correlation check with podium before each drop
Shared drives (2.6)	Removed 65 duplicate rows; flagged 175 rows `shared_drive = 1`	Needed for one row per driver per race

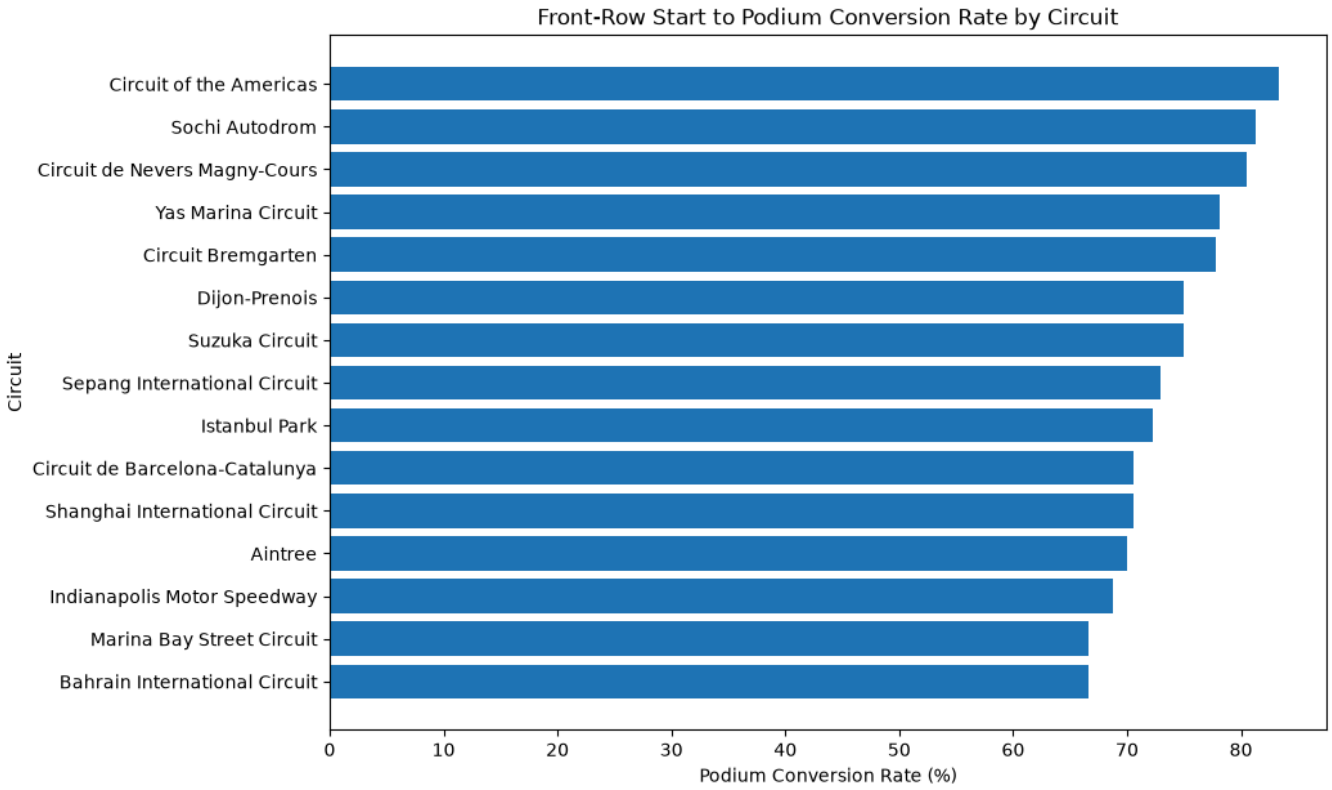
Correlation check before each column drop. Schedule columns in `races` and driver `number` / `code` correlate between −0.06 and +0.01 with podium, so they are both mostly empty and unrelated. The post-race `results` columns are strongly related (`position` −0.65), which is exactly why they are leakage and must go. For `q2` / `q3`, whether a value exists correlates up to −0.32 with podium, but the kept qualifying position already carries that signal more strongly (−0.49).

Duplicate rule (shared drives, mostly 1950s). When a driver appears twice in one race, we keep the row of the car they **finished in**, with that car's grid and result, and never mix columns from two rows.

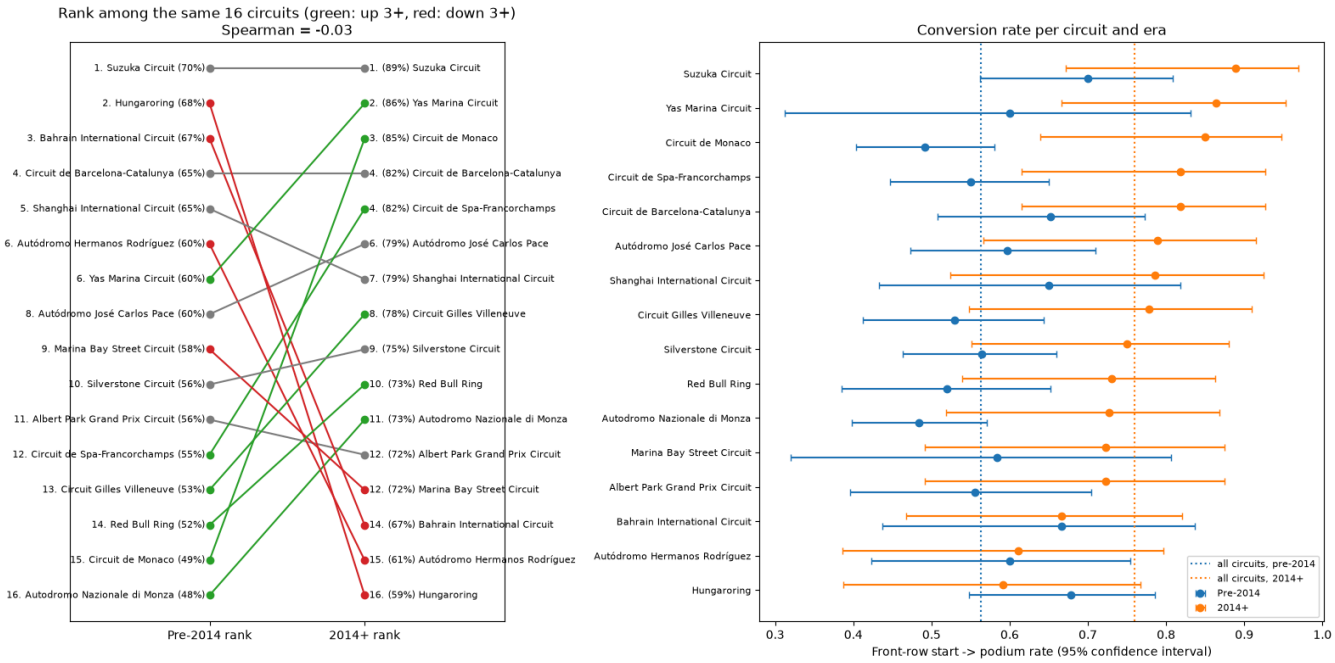
- **Takeover (44 pairs):** the driver's own car broke and they took over a teammate's car. We keep the taken-over car, the result they were credited with.
- **Swap (17 pairs):** drivers exchanged cars. The lowest `resultId` of a car belongs to its starter (true in 36 of 40 checkable cars), so the driver's later-added row marks the car they finished in.
- **One driver, two cars (1 pair):** Hernando da Silva Ramos, 1956 French GP; we keep his regular Gordini entry.

Question 1: front-row conversion by circuit

Answer. Among circuits with at least 5 races, Circuit of the Americas, Sochi and Magny-Cours convert a front-row start into a podium most often (above 80%). The ranking changes completely after 2014: over the 16 circuits with enough starts in both eras, the Spearman rank correlation is −0.03 ($p = 0.91$).



Does the ranking change after 2014? Yes. Only Suzuka and Barcelona stay in the top 5. Monaco rises from 15th to 3rd (49% to 85%), while the Hungaroring falls from 2nd to last (68% to 59%). Each circuit has only 10–30 front-row starts per era, so most single moves sit inside their 95% intervals; only Monaco's change is larger than its uncertainty. The safe conclusion is that the pre-2014 ranking does not carry over.



Why conversion rises after 2014. Front-row starts converted 56% of the time before 2014 and 76% from 2014. Most of the gap is reliability: 70% of front-row cars finished before 2014 against 92% after, while finishers converted at about the same rate (80% vs 83%). The rise is gradual over the decades, not a jump at the 2014 rules.

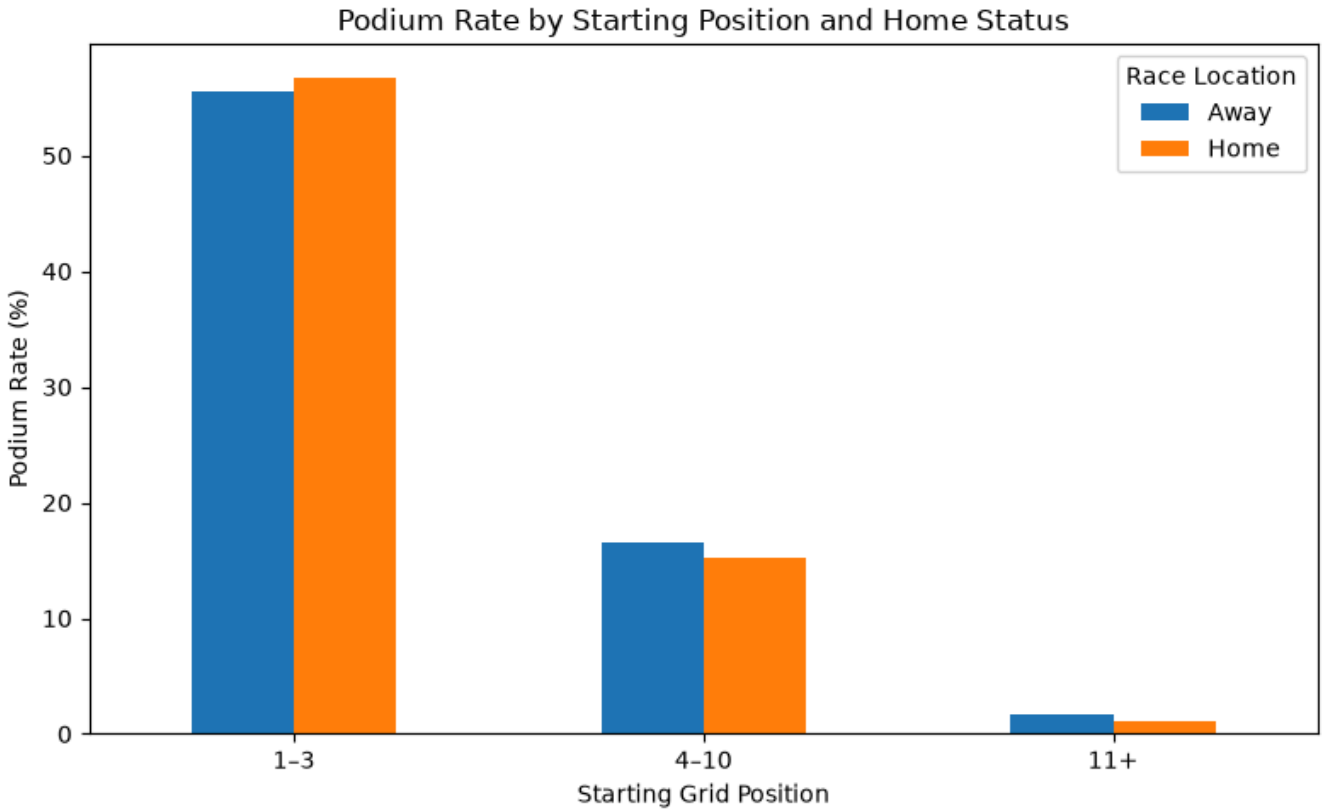
Qualifying vs grid (3.1.3). We define the front row by **grid**, the actual starting position after penalties. In 74 of 950 front-row starts with qualifying data the two differ. Drivers who qualified on the front row but started further back reached the podium 29% of the time, against 70% for those who started where they qualified. Ranking circuits by qualifying instead of grid gives a very similar order (Spearman 0.89).

Grain, missingness, cutoff. One row per driver per race; conversion = podiums ÷ front-row starts. `grid` has no missing values; qualifying exists only from 1994, so 3.1.3 uses races with qualifying records. The grid is known before the start; the finish is used only as the outcome.

Question 2: home-country podium effect

Answer. No. Once starting position is controlled for, drivers do not podium more often at home: the home-vs-away odds ratio is 0.96 (95% CI 0.81–1.14, p = 0.65).

Joins. `results` (grid, finish) → `drivers` (nationality) → `races` → `circuits` (country), plus our own `nationality_country` lookup, because nationality ("British") and country ("UK") use different names. `home = 1` when the circuit country matches any country of the driver's nationality.



Within each grid group the differences are small and point both ways: +1.2 points from grid 1–3 (56.8% home vs 55.6% away), −1.3 points from 4–10 and −0.5 points from 11+.

Test	Home vs away odds ratio	95% CI	p-value
Logistic regression, `podium ~ home + exact grid position`	0.96	0.81–1.14	0.65
Cochran–Mantel–Haenszel, 3 grid groups	0.93	0.79–1.10	0.42

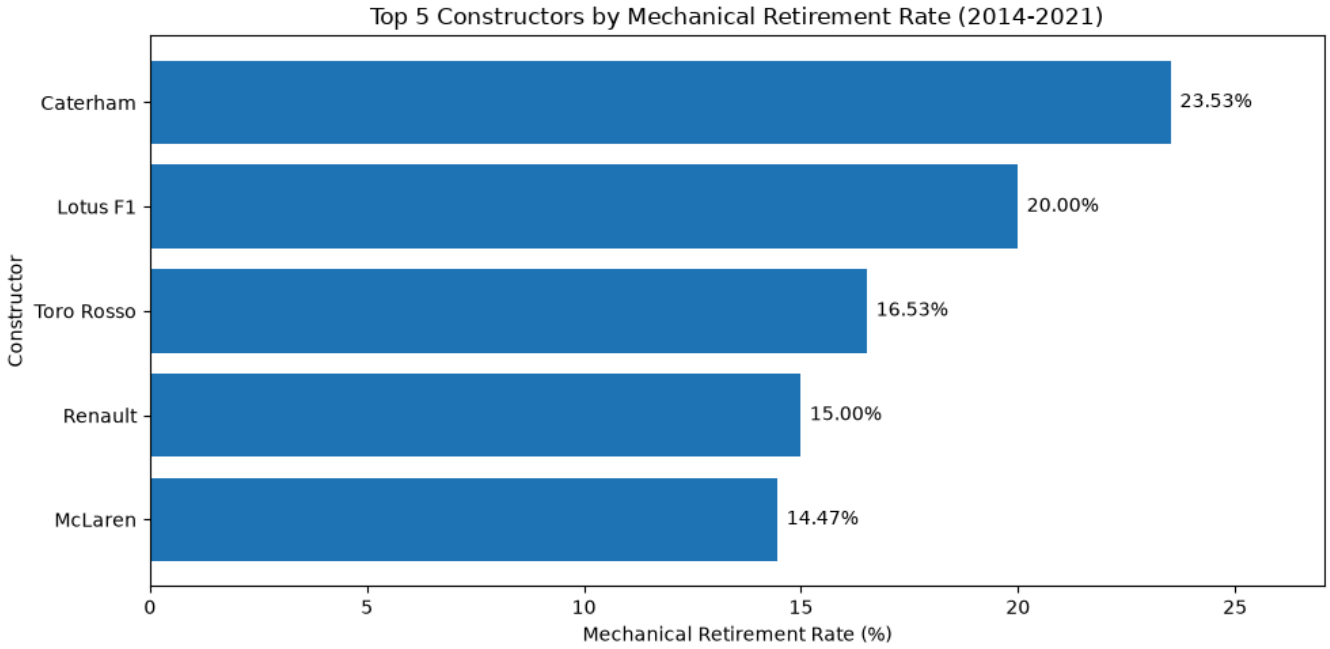
Both intervals include 1, and they cap any hidden home advantage at about +14% higher odds. This is an association, not a causal claim: `home` is based on nationality, and early decades include local one-off entries.

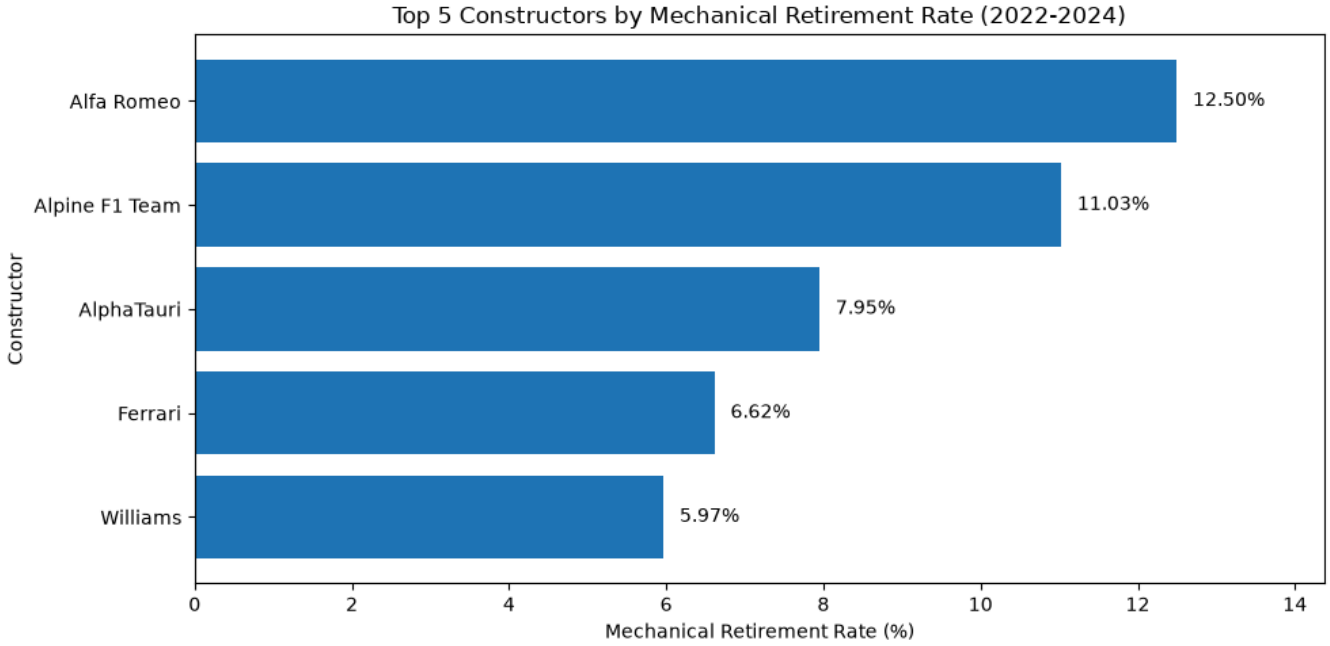
Grain, missingness, cutoff. One row per driver per race, with a valid grid position and a known home status. Home status and grid are known before the start; the finish is used only as the outcome.

Question 3: constructor mechanical-retirement rate

Answer. Caterham had the highest mechanical-retirement rate in 2014–2021 (23.5% of starts) and Alfa Romeo in 2022–2024 (12.5%). Across all teams the rate fell from 10.2% to 6.1% of starts.

Joins and definition. `results` → `status` (why the race ended) → `constructors`. We grouped every status value into mechanical (engine, gearbox, hydraulics, power unit, brakes and so on) or non-mechanical (accidents, driver errors, disqualifications). Rate = mechanical retirements ÷ race starts, per constructor and period.





- **Small samples:** Caterham (34 starts) and Lotus F1 (75) raced only part of 2014–2021, so their rates are less certain than teams with 200+ starts.
- **Team renames:** Toro Rosso → AlphaTauri, Renault → Alpine and Sauber → Alfa Romeo each get a new `constructorId`, so a change between periods can be the same team under a new name.
- **Sensitivity check (3.3.4):** a strict mapping that moves 26 debatable statuses (punctures, wing damage, out of fuel, stalls, wheel nuts) to non-mechanical keeps the same top 5 and the same leaders in both periods. The all-team drop holds (9.5% to 5.7%).

Grain, missingness, cutoff. One row per driver per race, 2014–2024; every status maps to a type, and withdrawn drivers are excluded from starts. This is a descriptive post-race analysis: `status` is never a model input.

Selected features and pre-race validity

The models use 26 features in 9 groups, each known after qualifying and before the start. Starting position is the strongest single signal (Spearman about −0.46 with podium), followed by form and standings (about 0.33–0.45).

Group	Features	Source tables	Why it is valid at the pre-race cutoff
Race context	`circuitId`, `circuit_country`	races, circuits	Fixed by the calendar
Starting grid	`grid_numeric`, `front_row`	results (grid only)	The grid is set before the start, after penalties
Qualifying	`qualifying_position`	qualifying	Qualifying happens before the race; filled with the grid before 1994

Group	Features	Source tables	Why it is valid at the pre-race cutoff
Driver attributes	`driver_age`, `driver_nationality`	drivers	Date of birth and nationality are fixed
Driver form	`driver_starts_before`, `driver_podium_rate_before`, last-5 and last-10 podium rate and average finish	results, earlier races only	Shifted one race back per driver (`shift(1)`)
Driver standings	`driver_points_before`, `driver_position_before`, `driver_wins_before`	driver_standings	Taken from the previous round of the same season
Constructor identity	`constructor_name`, `constructor_nationality`	constructors	The team is known when the car is entered
Constructor form	`constructor_races_before`, last-5 and last-10 podium rate and average finish	results, earlier races only	Combined per team per race first, then shifted one race back
Constructor standings	`constructor_points_before`, `constructor_position_before`, `constructor_wins_before`	constructor_standings	Taken from the previous round of the same season

Leakage check. For 500 random rows we recomputed the form and standings features from scratch, using only races held before that row's race. All 500 matched, and no feature correlates close to ±1 with podium, as a post-race column would.

No history yet. Before a driver's or team's first race of a season, points and wins are set to 0, because that is their real value. Championship position has no value before a race, so it stays missing and is filled with the training median plus a missing flag.

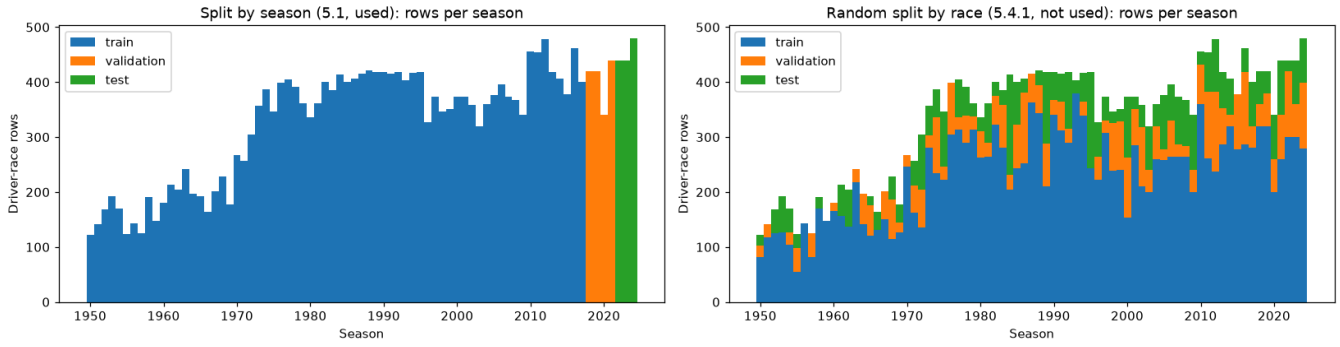
Excluded on purpose. Current-race `position`, `points`, `laps`, `time`, fastest-lap columns and `statusId` (post-race); `pit_stops`, `lap_times` and `constructor_results` (in-race or current-race); standings after the race; `sprint_results` (only from 2021); qualifying lap times (circuit-dependent, and `q2` / `q3` mostly empty); IDs and car numbers.

Removed after testing (6.1). Six built features are left out. `grid_vs_qualifying` and `has_quali_data` are almost always 0 or only flag data coverage; putting them back changes validation PR-AUC by under 0.003. `year`, `round`, `home` and `non_standard_start` are valid before the race but barely correlate with podium (|Spearman| ≤ 0.04). Leaving them out raises logistic regression's validation PR-AUC by 0.007 and leaves XGBoost's unchanged, so by our pre-set rule they go. They also cannot help on future seasons: every test `year` lies outside the training range, and section 3.2 found no home effect.

Models, metrics and experiments

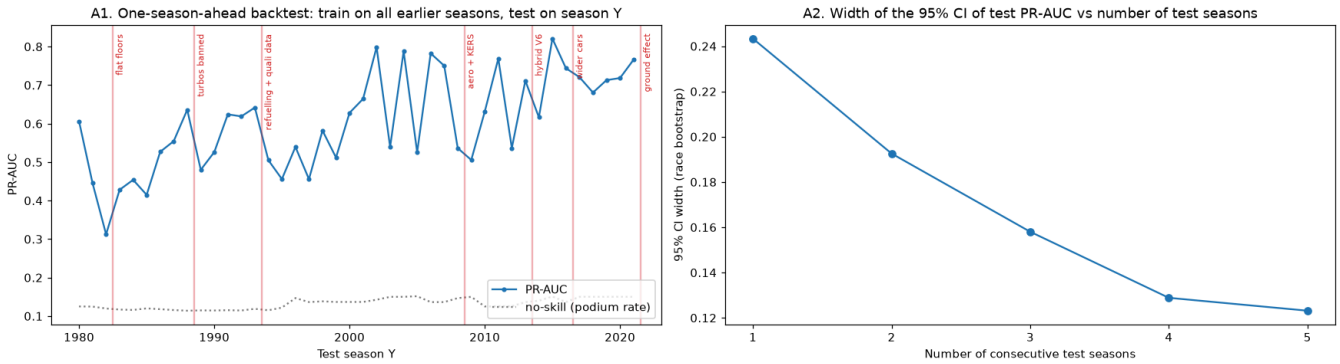
All four models beat the grid-only baseline on validation and are within noise of each other on the 68 test races (0.685–0.724 PR-AUC). XGBoost has the best validation PR-AUC of the statistical models (0.725) and is the final model.

Split by season. Training = 1950–2017 (965 races, 21,942 rows), validation = 2018–2021 (81 races, 1,620 rows), test = 2022–2024 (68 races, 1,359 rows). The model is meant to predict upcoming races, so every training race comes before every validation race, and every validation race before every test race. Preprocessing (median imputation with missing flags, scaling, one-hot encoding) is fitted on train only. Hyperparameters and F1 thresholds are chosen on validation; the test set is used only for reporting.



Why these seasons (5.1.1). Each boundary was tested in backtests inside 1950–2021, never on the test seasons:

Boundary	Choice	Evidence
Test	2022–2024	The full ground-effect era in the data. One test season gives a 95% interval of PR-AUC about 0.24 wide; three seasons cut it to 0.16
Validation	2018–2021	One season misleads (its score is off by about 0.10 from later seasons). From two seasons on no length is consistently better, so we take four for a more precise score (SD 0.035 instead of 0.050)
Training	1950–2017	Every start year from 1950 to 2006 scores within 0.015 of the others; starting in 2010 is 0.03 worse



Random split checked, not used (5.4.1). A random 70/15/15 split by race lets the model learn from the future: 100% of its test races share a season with training races, and 98% have training races from a later season. XGBoost retrained on it scores 0.595 test PR-AUC. It is lower only because its test races span 1950–2024 and older eras are harder (0.48 on 1950–1979 races, 0.73 on 2014–2024).

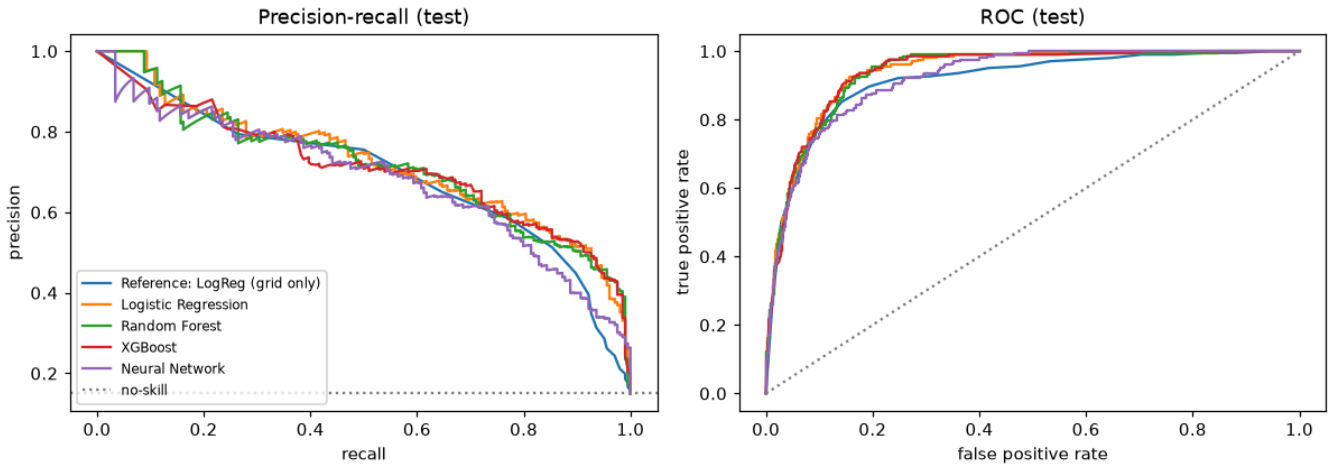
Metrics. PR-AUC is primary: podiums are rare (15% of test rows), and PR-AUC measures how precisely the model finds them (no-skill = 0.150 on the test set). ROC-AUC is secondary: threshold-free and widely reported, but it also rewards ranking the many easy back-of-grid cases, so it stays at 0.92–0.94 for every model. F1, precision and recall use the threshold that maximises F1 on validation.

Model	Val PR-AUC	Test PR-AUC	Val ROC-AUC	Test ROC-AUC	Val F1	Test F1	Val precision	Test precision	Val recall	Test recall
XGBoost (final)	0.725	0.706	0.939	0.938	0.726	0.679	0.663	0.642	0.802	0.721

Model	Val PR-AUC	Test PR-AUC	Val ROC-AUC	Test ROC-AUC	Val F1	Test F1	Val precision	Test precision	Val recall	Test recall
Neural network (shallow)	0.734	0.685	0.931	0.922	0.700	0.647	0.661	0.638	0.745	0.657
Random forest	0.720	0.718	0.941	0.937	0.719	0.668	0.656	0.630	0.794	0.711
Logistic regression (baseline)	0.713	0.724	0.936	0.939	0.720	0.680	0.642	0.615	0.819	0.760
Reference: grid only	0.668	0.647	0.918	0.914	0.712	0.649	0.712	0.650	0.712	0.647

Why XGBoost. The neural network leads on validation by only 0.009, far less than the noise of an 81-race validation score (SD about 0.035), so the models are tied. Among them we take XGBoost, the best statistical model on validation: it learns non-linear effects and can be explained exactly with TreeSHAP. The test set played no part; on test, logistic regression and random forest score a little higher (0.724 and 0.718), within noise.

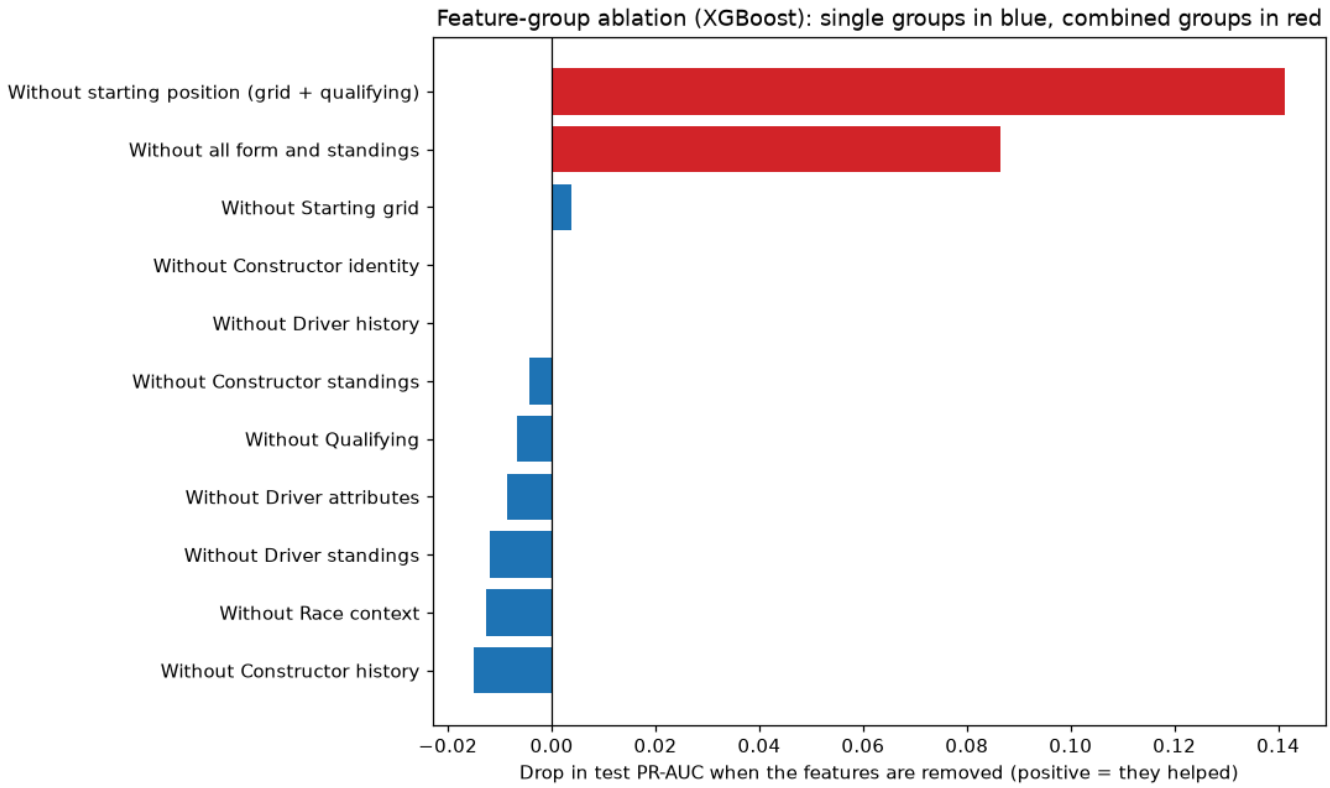
Training PR-AUC (0.57–0.64, notebook 5.4) is below validation for every model because the 1950–2017 training seasons are harder to predict, not because of underfitting. Early stopping keeps XGBoost at 17 shallow trees, so its probabilities are compressed (at most 0.44 on test) while its ranking is unaffected. The shallow network compared 1 layer of 32 units and 2 layers of 32-16 and 64-32, with dropout, L2 and early stopping.



Preprocessing-order experiments (5.5). Four controlled experiments with everything else fixed:

Experiment	Compared	Kept	Evidence
1. Impute vs scale order	impute → scale vs scale → impute	impute → scale	Same scores, but scale → impute leaves `driver_position_before` off-standard (mean about 0.01, sd 0.93)
2. Log transform	scaling only vs log of skewed counts, then scaling	scaling only	Skewness of points drops from about 5 to under 1, but PR-AUC changes by under 0.005
3. Missing qualifying fill	grid before preprocessing vs pipeline median	grid	Median scores +0.005 on validation, within noise; the grid is the real pre-1994 starting order
4. Class weighting	none vs balanced weights	none	Same or lower PR-AUC (XGBoost −0.007 on validation); weights inflate the mean predicted probability to 0.35–0.38 against a true rate of 0.15

Feature-group ablation (6). XGBoost retrained without each of the 9 groups. Removing any single group changes test PR-AUC by about 0.015 or less, because the groups overlap (grid and qualifying correlate 0.99). Removing grid and qualifying together drops test PR-AUC from 0.71 to 0.57; removing all four form and standings groups together drops it by 0.087.



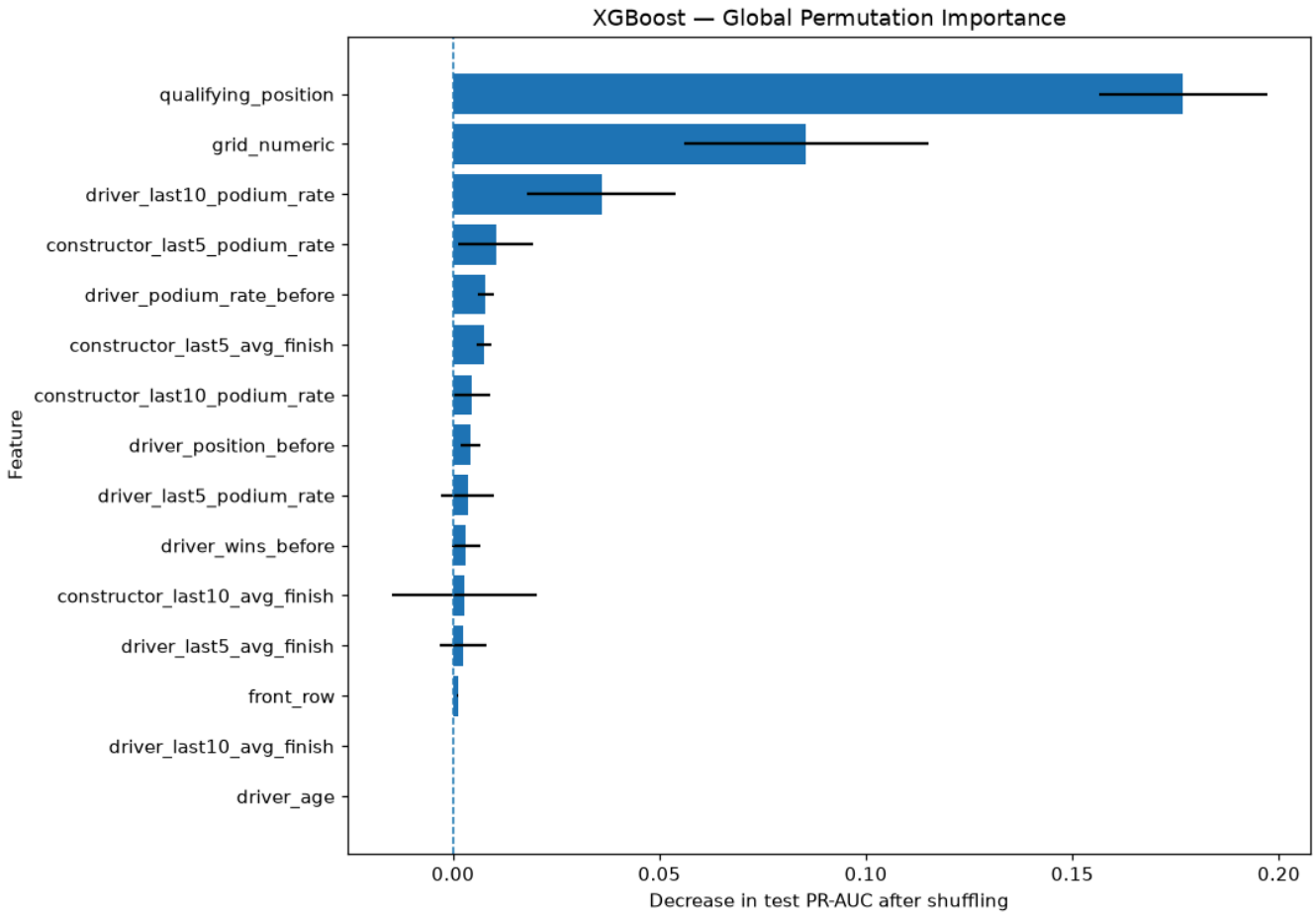
Testing the weakest features (6.1). Putting back `grid_vs_qualifying` and `has_quali_data` changes validation PR-AUC by under 0.003. Putting back `round`, `year`, `home` and `non_standard_start` lowers logistic regression's by 0.007 and leaves XGBoost's unchanged. Both sets stay out.

Explainability (XAI)

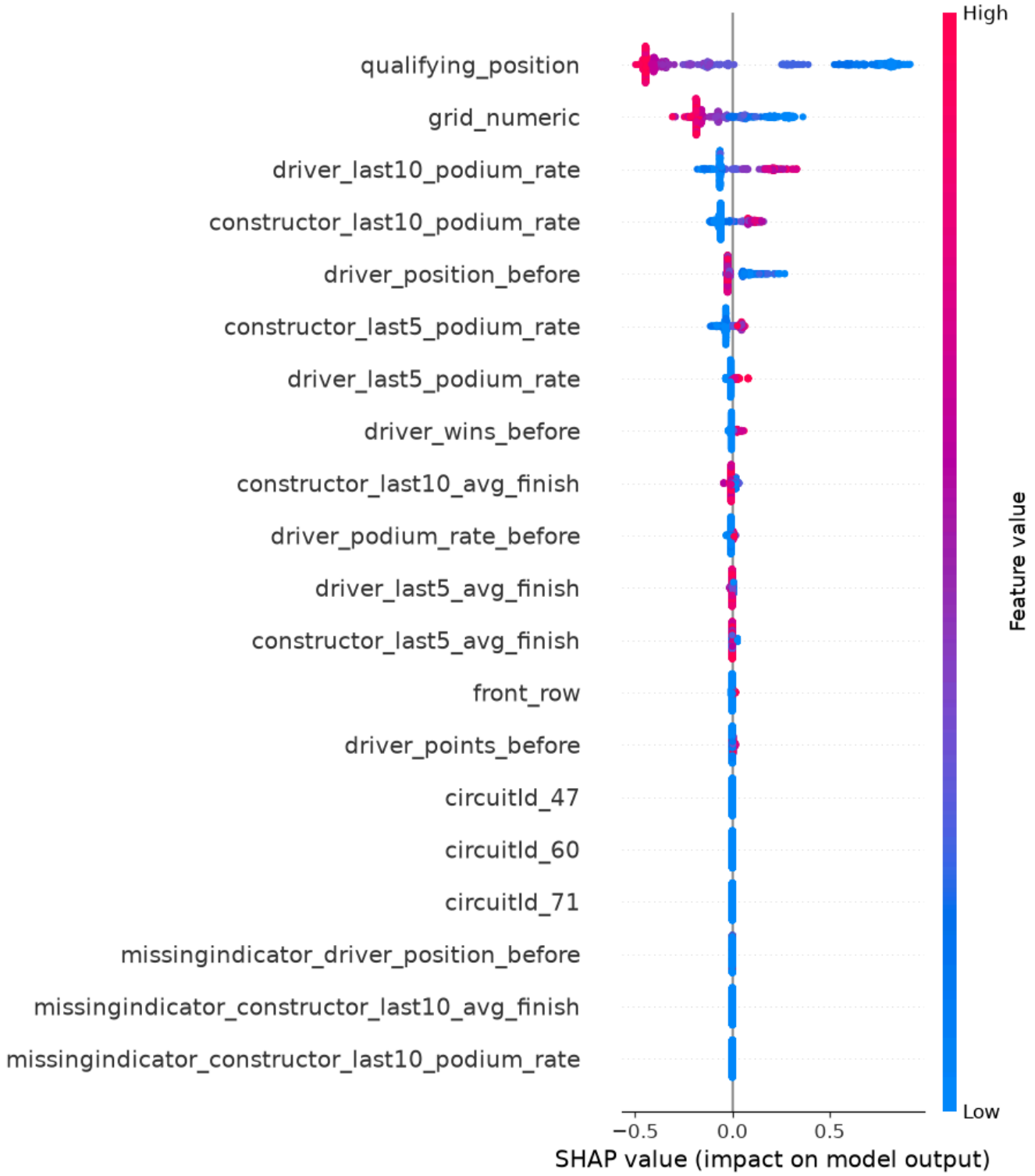
The final XGBoost model relies above all on starting position: on the 2022–2024 test set, shuffling `qualifying_position` costs about 0.18 PR-AUC and `grid_numeric` about 0.09, the driver's podium rate over the last 10 races about 0.04, and every other feature about 0.01 or less.

These explanations are not causal. A feature being important only means the model relies on it to make predictions. It does not prove the feature causes podium finishes: a good grid slot and a podium are both results of a fast car and driver.

Global: permutation importance. Each feature is shuffled 5 times on the test set and the drop in PR-AUC is measured. Many features score exactly 0 (team name, circuit, driver age): with only 17 shallow trees the model never splits on them. Values near zero are noise.

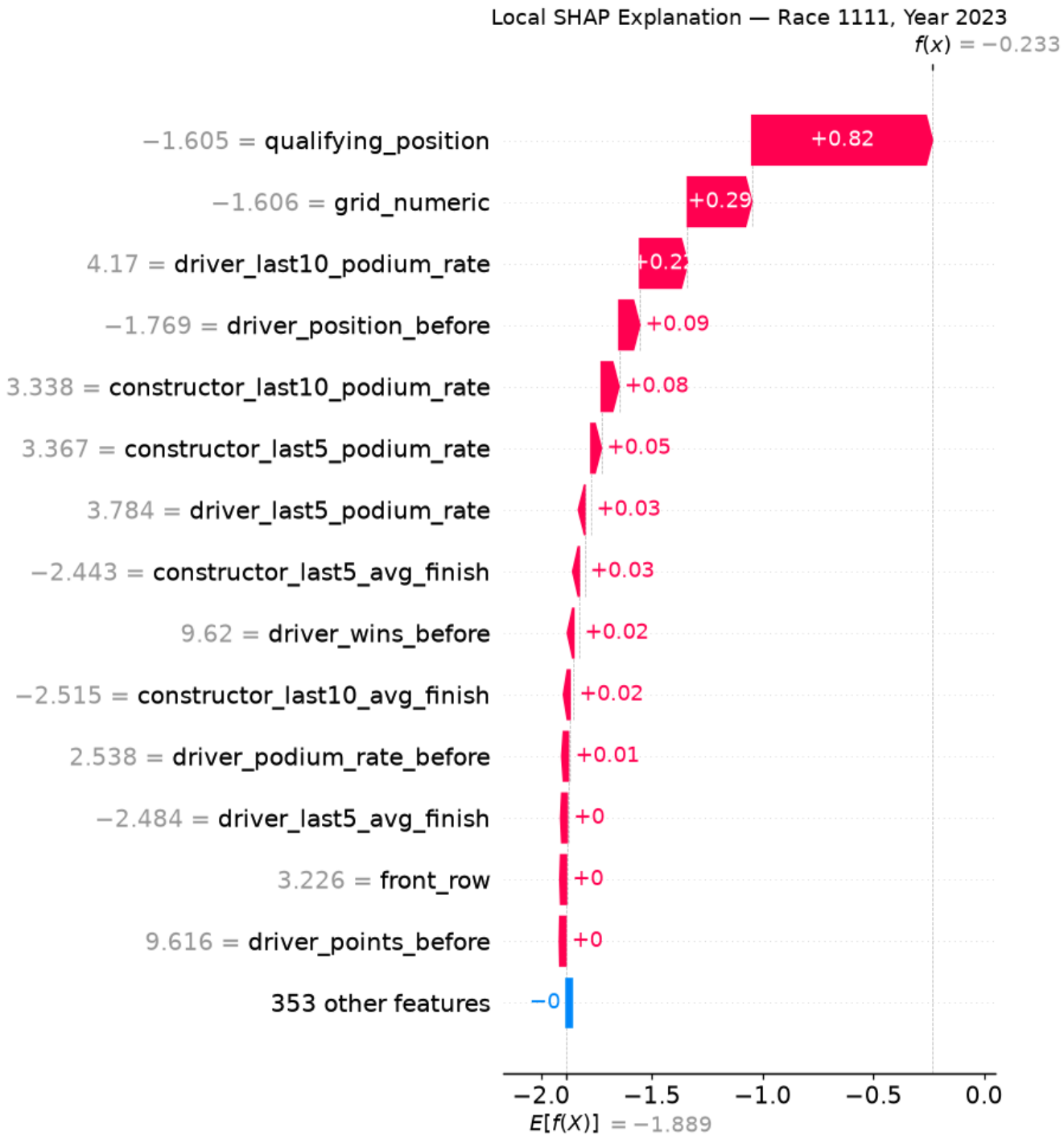


Global: SHAP. Starting position dominates, mostly through qualifying position (mean |SHAP| about 0.45 log-odds, against 0.15 for the grid): a front position pushes the prediction up by as much as +0.9 log-odds, a back position down by as much as −0.5. Next come the driver's and constructor's podium rates over the last 10 races (about 0.10 and 0.08): high recent podium rates push it up, a worse championship position before the race pushes it down.



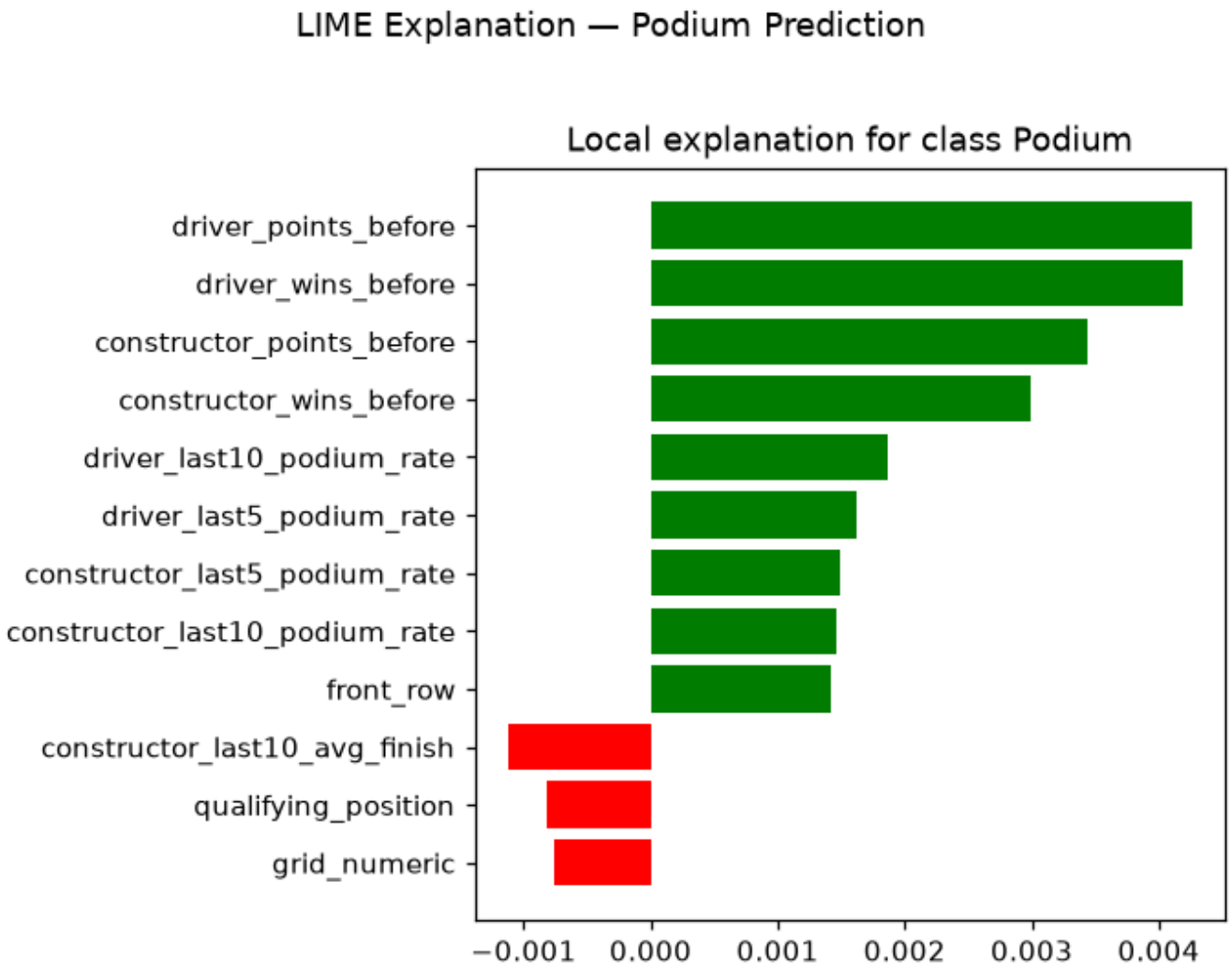
Local: Max Verstappen, Red Bull, 2023 Dutch GP (pole-sitter at his home race and championship leader; the highest predicted probability in the test set, about 44%; he won). The SHAP waterfall starts from the average prediction (about −1.9 log-odds, a 13% chance) and is pushed up mainly by pole position (qualifying +0.82, grid +0.29), a podium in each of his last 10 races (+0.22), his championship lead (+0.09) and Red Bull's recent

podium rates (+0.08, +0.05). 44% is far above the decision threshold (0.28), so the prediction is right; the probability itself is low because the 17-tree model's probabilities are compressed.



LIME fits a simple linear model around the same row. Its weights are tiny (at most about 0.004), because the compressed model's probability hardly changes near this row, but it agrees on the direction: a worse grid and qualifying position lowers the probability, more

points, wins and recent podiums raise it. LIME's weights come from random perturbations, so we treat it as a rough check and the agreement on starting position and form as the reliable part.



Inference function

`predict_podium(record)` returns a podium probability, a yes/no prediction and the list of missing features for one driver-race, and it still predicts when features are missing.

It uses the final XGBoost model and the F1 threshold chosen on validation. Missing numeric features are filled with the training median plus a missing flag, and missing categories are ignored, exactly as in training. This matters in practice: a rookie has no form or standings, and races before 1994 have no qualifying record.

Record (Lando Norris, 2024 Abu Dhabi GP, pole position)	Podium probability	Prediction	Missing features
Complete	0.407	Podium	none
Partially missing	0.111	No podium	8: qualifying position, driver standings (3), recent driver form (4)

He won the race. With every feature the probability (0.41) is well above the decision threshold (0.28), so the prediction is right; the value itself is low because XGBoost's probabilities are compressed. Without qualifying, standings and form, the model falls back on training medians that describe an average driver, so its probability drops: with less information it is less confident, which is the intended behaviour.

Known limitations

The biggest limits are the small test set (68 races, one regulation era) and the large changes between F1 eras.

- **Small validation and test sets.** 81 validation races and 68 test races: PR-AUC differences between models below about 0.05 are within noise, so the four models are statistically tied on test, and XGBoost is chosen on validation and explainability, not a clear win.
- **One era tested.** The test score describes the 2022–2024 ground-effect era. Older eras are much harder to predict (XGBoost about 0.48 PR-AUC on 1950–1979 races vs 0.73 on 2014–2024 in the random-split check), and a future rule change could lower the score again.
- **Compressed probabilities.** Early stopping keeps XGBoost at 17 trees, so its probabilities stay close to the base rate (at most 0.44 on test). Rankings are good, but a reported probability understates a front-runner's chance; yes/no uses the validation F1 threshold (0.28).
- **Qualifying-coverage drift.** Qualifying records exist only from 1994, so for about 58% of rows the qualifying position is filled with the starting grid. Experiment 3 shows the fill barely changes the scores, but qualifying adds nothing beyond the grid for older races.
- **Era sensitivity.** Field sizes, reliability, points systems and regulations changed a lot. Front-row conversion was about 46–50% in the 1960s–80s against 75% in the 2010s–20s, and mechanical retirements fell from 10.2% to 6.1% of starts after 2021.

- **Shared drives (1950s).** 119 rows share a finishing position with another driver, all flagged `shared_drive`. The swap rule held in 36 of 40 checkable cars, so a few of the 17 swap decisions may be wrong.
- **Withdrawn drivers kept.** 335 rows (1.3%) are drivers who withdrew; they stay with grid 0, which `grid_numeric` places behind the last normal starter.
- **Missing history.** Points and wins before a season's first race are correctly 0, but about 3,000 rows have no championship position before the race and newcomers have no form; these are filled with training medians, which describe an average driver.
- **Tuning.** Hyperparameter grids are small, and early stopping uses the validation set, so validation scores are slightly optimistic; the test set is the fair comparison.
- **Team identity.** A renamed team (Toro Rosso → AlphaTauri, Renault → Alpine) gets a new `constructorId`, so its form history restarts and a brand-new team has no one-hot history.
- **Explanations are not causal**, and LIME varies between runs because it uses random perturbations.